EFFECTIVITY OF DESCENT DATA FOR (QUASI-)PROJECTIVE SCHEMES OVER A FINITE LOCALLY FREE COVER

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Lemma lemma-descend-along-finite (canonical tag 0CCI). cited at proof lines 5299-5300. The criterion on affine opens through finitely many points.
- Morphisms lemmas 07RL, 01VW, 01VH, 01VN. cited at proof lines 5302-5321. Chain of reductions among the five cases.
- Lemma properties-lemma-ample-finite-set-in-affine (01ZY). cited at proof lines 5307-5310. Finitely many points land in an affine open when an ample sheaf exists.

Source reference. Original lemma 0CCI: sources/stacks-chow/tags.txt, source lines 15508–15508. Cited at proof lines 5299-5300 as the criterion being verified; load-bearing.

Source reference. Original lemma 07RL: sources/stacks-chow/tags.txt, source lines 9871–9871. Cited at proof lines 5302-5304 for case (1); load-bearing.

Source reference. Original definition 01VW: sources/stacks-chow/tags.txt, source lines 2461–2461. Cited at proof lines 5304-5306 for case (2); load-bearing.

Source reference. Original lemma 01ZY: sources/stacks-chow/tags.txt, source lines 2613–2613. Cited at proof lines 5307-5310 for case (3); load-bearing.

Source reference. Original definition 01VH: sources/stacks-chow/tags.txt, source lines 2446–2446. Cited at proof lines 5314-5316 for case (4); load-bearing.

Source reference. Original lemma 01VN: sources/stacks-chow/tags.txt, source lines 2452–2452. Cited at proof lines 5317-5321 for case (5); load-bearing.

Lemma 0.2. *Let $X \to Y$ be a surjective finite locally free morphism. Let V be a scheme over X such that one of the following holds*

(1) *$V \to X$ is projective,*
(2) *$V \to X$ is quasi-projective,*
(3) *there exists an ample invertible sheaf on V,*
(4) *there exists an X-ample invertible sheaf on V,*
(5) *there exists an X-very ample invertible sheaf on V.*

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Then any descent datum on $V/X/Y$ is effective.

Proof. We check the condition in Lemma 0CCI. Let $y \in Y$ and $v_1, \ldots, v_d \in V$ points over y. Case (1) is a special case of (2), see Morphisms, Lemma 07RL. Case (2) is a special case of (4), see Morphisms, Definition 01VW. If there exists an ample invertible sheaf on V, then there exists an affine open containing $v_1, \ldots, v_d$ by Properties, Lemma 01ZY. Thus (3) is true. In cases (4) and (5) it is harmless to replace Y by an affine open neighbourhood of y. Then X is affine too. In case (4) we see that V has an ample invertible sheaf by Morphisms, Definition 01VH and the result follows from case (3). In case (5) we can replace V by a quasi-compact open containing $v_1, \ldots, v_d$ and we reduce to case (4) by Morphisms, Lemma 01VN. □